

HealthConnect Clinic Experience Lab

Week 5 Project Summary - Data Analytics

1. What I Planned to Accomplish

For Week 5, I planned to move from the problem-understanding stage, which I completed in Week 4, into a deeper analysis of the HealthConnect appointment data. My focus was to examine no-show patterns and calculate the proposed KPIs as well as identify factors that could be associated with missed appointments.

2. What I Completed

In this project, I created a separate working copy of the dataset so that the original data remained unchanged. I also reviewed the data quality, carried out exploratory analysis and used PivotTables in Excel to compare appointment outcomes across different factors.

I also developed a Power BI dashboard with the following:

- a) Total Appointments
- b) No-Show Rate
- c) Average Booking Lead Time for No-Shows
- d) Repeat No-Show Rate
- e) Reminder No-Show Rate
- f) No-Show Rate by Appointment Time
- g) No-Show Rate by Appointment Day
- h) No-Show Rate by Distance to Clinic
- i) No-Show Rate by Waiting Time
- j) Appointment Type and Gender slicers

3. Key Findings

The analysis showed that 48.46% of the 5,000 appointments were no-shows. Appointments where reminders were sent had a lower no-show rate than appointments without reminders.

Previous no-show history was one of the strongest patterns identified. The repeat no-show rate among patients with at least one previous no-show was 55.41%.

No-show appointments also had an average booking lead time of 34.53 days, compared with 24.52 days for attended appointments.

4. Major Challenges Encountered

One challenge was making sure that the Power BI measures were calculating the correct values. I had to check the underlying counts and correct some calculations before finalizing the dashboard.

I also encountered blank values in some dashboard slicers and resolved this by filtering out the blank categories.

Another challenge I identified during data preparation was inconsistency in the date fields. What I did was to retain the supplied booking_lead_days variable instead of calculating lead time from the dates.

5. Important Decisions Made

I retained the missing values in the distance-to-clinic and waiting-time fields rather than just replacing them with assumed values. There were 90 missing distance values and 60 missing waiting-time values.

I also grouped distance and waiting-time values into ranges in Power BI so that their patterns could be presented more clearly.

6. Changes to My Week 4 Approach

There was no major change to the overall direction of the project. The Week 5 work continued from the business questions and KPIs I developed during Week 4. The main difference was that Week 5 moved from planning into actual analysis and dashboard development.

7. Cross-Track Collaboration

I did not complete a formal interaction or collaboration with another HealthConnect project track during Week 5 project.

8. Remaining Work

The next stage for me is to refine the analysis where necessary. I will document the findings clearly and continue developing the HealthConnect project based on the patterns I identified during the Week 5 project.

9. Proposed Focus for Week 6

For Week 6, I plan to build on the Week 5 findings by refining the dashboard and reviewing the strongest no-show patterns, while preparing the analysis for the next stage of the HealthConnect Experience Lab.